



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-2024
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: ENVIRONMENTAL APPLICATIONS [SET-A]

FULLMARKS: 100
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of five printed pages.*

SECTION A (40 Marks)
Attempt all questions from this Section

Question 1

Choose the correct answers to the questions from the given options. [20]
(Do not copy questions, write the correct answers only).

- i. Hydro energy can be produced from
 - a) Rainwater
 - b) Perennial river
 - c) Stored well water
 - d) Stagnant water
- ii. A soil conservation strategy by placing boulders and sand bags on the path of water flow is
 - a) Basin dam
 - b) Contour dam
 - c) Check dam
 - d) Big dam
- iii. Community forestry benefits
 - a) Individuals
 - b) Local community
 - c) Forest officials
 - d) Farmers
- iv. Any system that processes waste before discharging it to the environment is
 - a) Screening
 - b) Sorting
 - c) EOP treatment
 - d) Landfill

- v. The vehicle which uses two or more power sources is a
- Hybrid vehicle
 - Electric vehicle
 - Motorized vehicle
 - High voltage vehicle
- vi. Assertion: EIA is based on grass root upward planning.
Reason: Higher authorities and Government officials are involved in the assessment stage of EIA.
Choose the correct option
- A is true and R is the correct reason for A.
 - A is true but R is not the correct reason for A.
 - A is true but R is false.
 - A is false but R is true.
- vii. Overgrazing can be controlled by
- Stall feeding
 - Open grazing
 - Rearing a big herd
 - Not keeping any herd
- viii. The Montreal Protocol is an international treaty designed to control
- Carbondioxide emissions
 - loss of biodiversity
 - ozone depletion
 - soil erosion
- ix. Study the following picture and answer the question:



The device converts light energy to-

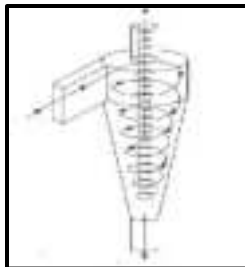
- mechanical energy
 - wind energy
 - electrical energy
 - chemical energy
- x. With reference to NRA consider the following statements
- It includes cost of labour
 - Depreciation cost is not considered
 - It includes cost of machines
 - Exploration costs are not part of NRA
- Choose the correct option:
- 1 and 2
 - 1,2 and 4
 - 1 and 3
 - 2 and 4

- xi. Madhav owns two large farms and 3 small farms while his brother Jadu works as labourer in any of the above mentioned farms as per his wish. This is an example of
- Inter generation equity
 - Intra generation equity
 - Intra generation inequity
 - Intergeneration inequity
- xii. Accumulating and storing water for future use is called
- Rainwater harvesting
 - Water harvesting
 - Water management
 - Rain water conservation
- xiii. The area where rainwater falls is known as
- Catchment
 - Conduit
 - Drainage
 - Aquifer
- xiv. Raghurajpur is famous for “Patachitra” which is cloth based painting. This can be a part of
- PETA
 - Gandhigram
 - Dastkar
 - Auroville
- xv. Child labour law prohibits the employment of children below age
- 12
 - 14
 - 18
 - 10
- xvi. Remote sensing cannot be used for
- Predicting weather
 - Locate active mining sites
 - Calculate NRA
 - Drought warnings
- xvii. If SPM in the air is high we will suffer from
- Stomach infection
 - Cardiac arrest
 - Respiratory diseases
 - Liver cirrhosis
- xviii. Khadi cloth fabric is
- Cotton
 - Chiffon
 - Crepe
 - Nylon
- xix. Fuel wood is _____
- A form of fossil fuel
 - A solid fuel biomass
 - A form of liquid fuel
 - A form of alternative fuel
- xx. World Environment Day is celebrated on
- 6th June
 - 14th June

- c) 5th June
- d) 5th July

Question 2

- a) Define shelter belts. How are they useful? [2]
- b) Mention two advantages of brain drain. [2]
- c) Study the picture. [3]



Identify and state the working principle of the device.
Mention the specific use of the device.

- d) State three initiatives you can take as a student to conserve water. [3]

Question 3

- a) Mention two advantages of GDP. [2]
- b) What is a catalytic converter? [2]
- c) Examine the role of law in controlling air pollution. [3]
- d) Who are referred to as northern countries? Mention any two exceptions. What is their view point on environmental degradation? [3]

SECTION B (60MARKS) (Attempt any four questions from this section)

Question 4

- a) Write the pros and cons of using chemical fertilizers. [5]
- b) 'EIA is an exercise to be carried out before any project or major activity undertaken'. How is it beneficial? [5]
- c) 'Auroville is an ideal township showcasing sustainable development'. Comment. [5]

Question 5

- a) How do traditional fishing techniques differ from modern ones? [5]
- b) Who is responsible for village or small town level administration in India? How do they contribute in development of these areas? [5]
- c) How do NGO s help in women welfare? [5]

Question 6

- a) How is globalization responsible for environmental degradation? [5]
- b) Wetlands are vital for human survival. List any three importance of wetlands. Name the international convention for protecting wetlands. Where was the convention adopted? [5]
- c) Write the first 5 steps of EIA. [5]

Question 7

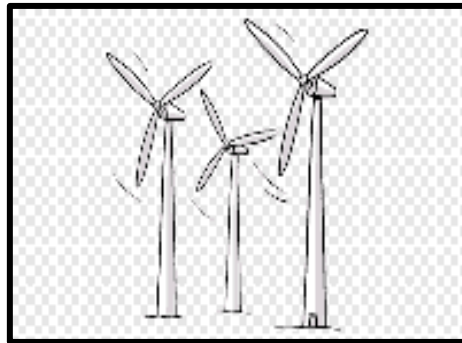
- a) Why environmentalists do not prefer large dams? [5]
- b) A community in Tamil Nadu wanted to conserve water for the benefit of the society. They came up with a particular plan of action which is in sync with the place's traditions.
 - i. What according to you is their plan? Give a rough sketch of the structure.
 - ii. Mention any 3 benefits expected out of this plan. [5]
- c) Write any 5 consequences of soil erosion. [5]

Question 8

- a) Enlist 5 eco friendly activities that we can follow in our daily life. [5]
- b) Explain Malthus' theory on population. [5]
- c) If we do not take care of the air, water and soil, what impacts will it have on human health?
Mention any 2 problems that will add up if wastes are not properly disposed off. [5]

Question 9

- a) Study the picture and answer the following questions: [5]



- i. Identify the picture.
- ii. What are the individual devices known as and state their function.
- iii. Name the largest of these which is operational onshore. Where is it found in India?
- b) What is a dry compost toilet? How does it benefit the people of Ladakh? [5]
- c) Write a short note on Basel Convention. [5]

